

8. Hydrogen peroxide solution, H_2O_2 , is used in commercial stain removers.

A GCSE class investigated how effective four stain removers are at removing stains. Stain removers **A**, **B**, **C** and **D** contain different concentrations of hydrogen peroxide.

The students tested how effective each one is at removing an identical oil stain from four towels.

Their findings are outlined below.

Stain remover A

Working temperature	= 50 °C
Cost per 100 cm ³	= 99p
Time to remove stain	= 40 min
Volume needed	= 20 cm ³

Stain remover B

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.99
Time to remove stain	= 40 min
Volume needed	= 10 cm ³

Stain remover C

Working temperature	= 20 °C
Cost per 100 cm ³	= £2.49
Time to remove stain	= 20 min
Volume needed	= 5 cm ³

Stain remover D

Working temperature	= 30 °C
Cost per 100 cm ³	= £1.49
Time to remove stain	= 30 min
Volume needed	= 10 cm ³

- (a) In carrying out this investigation, which variables were kept the same in order to get valid results? Tick (✓) the correct answer. [1]

type of oil used, towel material and volume of hydrogen peroxide

☐

type of oil used, towel material and temperature of stain remover

☐

type of oil used and towel material

☐

type of oil used, towel material and cost of stain remover

☐


- (b) Tick (✓) **all** of the statements which **could** explain why stain remover **A** has to be heated to 50 °C before it removes the stain. [1]

it is the cheapest stain remover

☐

it is heat resistant

☐

it has a low concentration of hydrogen peroxide

☐

it takes a long time to work

☐

- (c) The students found that stain removers **B** and **D** used the same volume and worked best at the same temperature.

Assuming that they have the same hydrogen peroxide concentration, suggest a possible reason why **D** removes the stain more quickly than **B**. [1]

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(d) One student went on to investigate the decomposition of hydrogen peroxide.

The equation for the reaction is as follows.

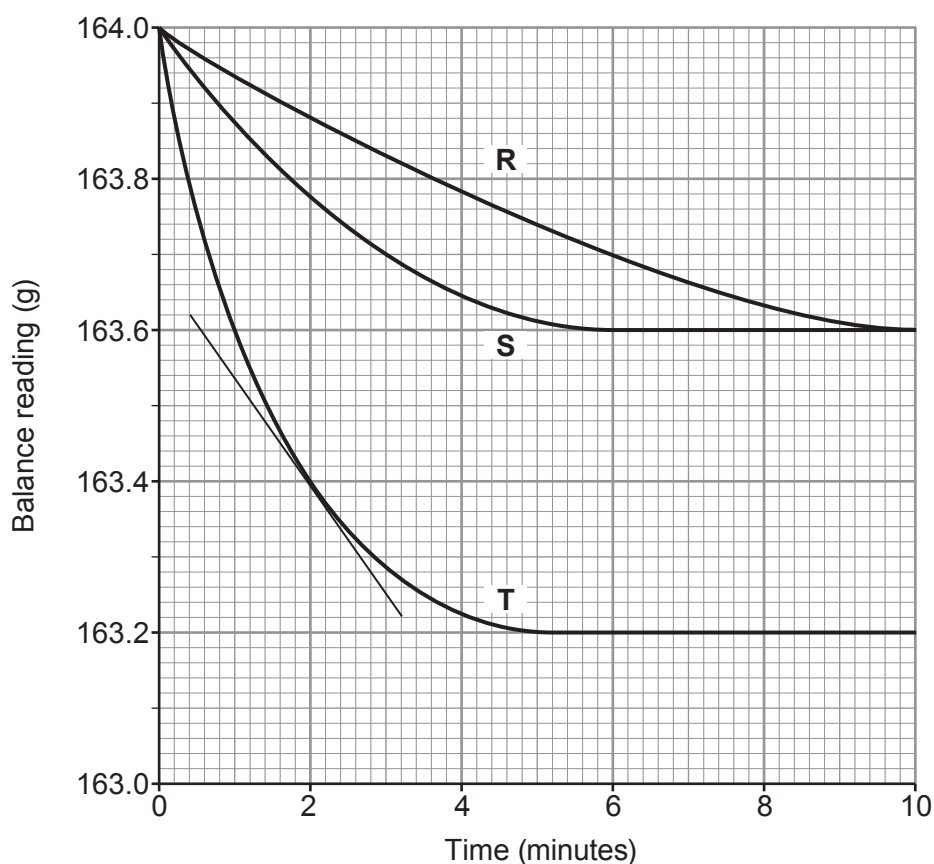


The student investigated the effect of changing the concentration of the hydrogen peroxide solution on the rate of the reaction. She used manganese dioxide as a catalyst in each experiment.

This is the method she used.

- Pour 50 cm³ of hydrogen peroxide solution of concentration **R** into a conical flask on a digital balance.
- Add 1 g of catalyst and place some cotton wool loosely in the neck of the flask. Record the balance reading and immediately start a stopwatch.
- Record the balance reading every minute until the mass no longer changes.
- Carry out the experiment twice more using hydrogen peroxide of different concentrations, **S** and **T**.

Her results are plotted on the grid below.



- (i) Using the tangent shown on the graph, calculate the rate of reaction for concentration **T** at 2 minutes. Show your working. [2]

Rate at 2 minutes = g/minute

- (ii) The initial rate for concentration **S** is half the initial rate for concentration **T**. Explain this difference in rate in terms of the particle theory. [3]

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END OF PAPER

